

D582

Introduction to Statistics for Research

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A patient survey used by a 12-clinic network, which recently switched survey vendors, asks patients to rate their care as poor, fair, good, or excellent. What level of measurement is this rating?

- A. Ratio, because each rating can be coded 1 to 4.
- B. Nominal.
- C. Ordinal.
- D. Interval.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A neighborhood survey collected by volunteers recorded these household incomes, in thousands of dollars: 38, 42, 45, 47, 50, 52, 410. Which measure best describes a typical household?

- A. The range, \$372,000.
- B. The mean, about \$97,700.
- C. The median, \$47,000.
- D. \$410,000.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A licensing exam given at 40 testing sites has a mean of 500 and a standard deviation of 100. A candidate scored 650. Her advisor noted she studied for 6 weeks. What is her z-score?

- A. 1.5
- B. -1.5
- C. 150
- D. 6

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Scores on a quiz have a mean of 70 and a standard deviation of 8. What raw score has a z-score of -1.25 ?

- A.** 68.75
- B.** 80
- C.** 60
- D.** 8

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A trial across three hospitals, sponsored by the dressing's manufacturer, compared healing times for a new wound dressing and standard care. The researchers set $\alpha = .05$ and found $p = .03$. What is the correct statistical decision?

- A.** Fail to reject the null, because p is larger than α .
- B.** The dressing works for 97% of patients.
- C.** Discard the results, since the manufacturer paid for the study.
- D.** Reject the null hypothesis; the difference is statistically significant.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Researchers conclude a tutoring program improves scores and reject the null hypothesis, but in truth the program has no effect. What kind of error is this?

- A.** No error, because rejecting the null guarantees a real effect.
- B.** Type II error.
- C.** Sampling bias.
- D.** Type I error.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

An analyst at a state health agency, which just launched a new data dashboard, wants to know whether mean systolic blood pressure differs among four diet groups: vegan, vegetarian, pescatarian, and omnivore. Which test fits?

- A. One-way ANOVA.
- B. Chi-square test of independence.
- C. Pearson correlation.
- D. Independent-samples t-test.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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8. [TRANSFER — reversed stem / different variable]

Which research question is best answered with a chi-square test of independence?

- A. Do mean scores differ across three teaching methods?
- B. Is vaccination status (yes or no) related to region (north, south, east, west)?
- C. Do men and women differ in mean cholesterol?
- D. Does weekly study time predict exam score?

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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9.

In a study of 250 adults recruited through a fitness app, the correlation between daily steps and resting heart rate is $r = -.60$. Which interpretation is correct?

- A. A weak relationship, because the value is negative.
- B. Steps explain 60% of the variation in resting heart rate.
- C. Walking more causes a lower resting heart rate.
- D. A moderately strong negative relationship; steps share about 36% of the variation in resting heart rate.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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10.

A hospital unit, which renovated its lobby last spring, found that predicted weekly overtime hours = $12 + 0.8 \times (\text{weekly admissions})$. How many overtime hours are predicted for a week with 30 admissions?

- A.** 360
- B.** 36
- C.** 42
- D.** 24

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

11. [TRANSFER — reversed stem / different variable]

Using predicted overtime hours = $12 + 0.8 \times (\text{weekly admissions})$, how many admissions would predict 44 overtime hours?

- A.** 12
- B.** 40
- C.** 55
- D.** 47.2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

12.

Birth weights at a regional hospital that delivers about 4,000 babies a year are roughly normal, with a mean of 3,400 g and a standard deviation of 500 g. About what percentage of babies weigh between 2,400 g and 4,400 g?

- A.** About 95%
- B.** About 47.5%
- C.** About 68%
- D.** About 99.7%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A patient survey ~~used by a 12 clinic network, which recently switched survey vendors,~~ asks patients to rate their care as poor, fair, good, or excellent. What level of measurement is this rating?

Target word: level of measurement **Decides:** Ordered categories without equal, measured gaps.

A	Dropped element	Coding with numbers does not create equal gaps or a true zero.
B	Adjacent category	Nominal categories have no order; these do.
C	Correct	The categories have a clear order, but the gaps between them are not equal units.
D	Confusion set	Assumes the gap from poor to fair equals the gap from good to excellent.

2. Answer C Concept 2

A neighborhood survey ~~collected by volunteers~~ recorded these household incomes, in thousands of dollars: 38, 42, 45, 47, 50, 52, 410. Which measure best describes a typical household?

Target word: best center for skewed data **Decides:** One extreme value (410) skews the mean.

A	Adjacent category	$410 - 38$ measures spread, not center.
B	Confusion set	$684 / 7 = 97.7$; pulled far above six of seven households by the outlier.
C	Correct	The middle of seven ordered values is 47; it resists the outlier.
D	Wrong stem number	The outlier itself.

3. Answer A Concept 3

A licensing exam ~~given at 40 testing sites~~ has a mean of 500 and a standard deviation of 100. A candidate scored 650. ~~Her advisor noted she studied for 6 weeks.~~ What is her z-score?

Target word: z-score **Decides:** $(650 - 500) / 100$.

A	Correct	$(650 - 500) / 100 = 1.5$.
B	Inverted operation	Subtracts in the wrong order; she scored above the mean.
C	Dropped element	Subtracts the mean but never divides by the SD.
D	Wrong stem number	The weeks of study, which do not decide the z-score.

4. Answer C Concept 3 • transfer

Scores on a quiz have a mean of 70 and a standard deviation of 8. What raw score has a z-score of -1.25 ?

Target word: solve for the raw score **Decides:** $x = \text{mean} + z \times \text{SD}$.

A	Dropped element	Subtracts 1.25 without multiplying by the SD.
B	Inverted operation	Adds 10 instead of subtracting; a negative z is below the mean.
C	Correct	$70 + (-1.25 \times 8) = 70 - 10 = 60$.
D	Wrong stem number	The standard deviation, not a score.

5. Answer D Concept 4

A trial ~~across three hospitals, sponsored by the dressing's manufacturer,~~ compared healing times for a new wound dressing and standard care. The researchers set $\alpha = .05$ and found $p = .03$. What is the correct statistical decision?

Target word: decision at $\alpha = .05$ **Decides:** $p = .03$ is less than $\alpha = .05$.

A	Inverted operation	.03 is smaller than .05.
B	Confusion set	p is not the chance the treatment works or the share of patients helped.
C	Camouflage	Funding source is a bias concern to review, but it does not change the statistical decision.
D	Correct	$p \leq \alpha$ leads to rejecting the null.

6. Answer D Concept 4 • transfer

Researchers conclude a tutoring program improves scores and reject the null hypothesis, but in truth the program has no effect. What kind of error is this?

Target word: classify the error **Decides:** A true null was rejected.

A	Absolute language	Significance tests always carry an α risk of false positives.
B	Inverted operation	Type II is failing to reject a false null, a false negative.
C	Adjacent category	A flaw in who was sampled, not a decision error.
D	Correct	Rejecting a true null is a false positive, a Type I error.

7. Answer A Concept 5

An analyst at a state health agency, ~~which just launched a new data dashboard,~~ wants to know whether mean systolic blood pressure differs among four diet groups: vegan, vegetarian, pescatarian, and omnivore. Which test fits?

Target word: test choice **Decides:** One continuous outcome compared across four groups.

A	Correct	Compares means across three or more groups in one test.
B	Confusion set	For two categorical variables; blood pressure is continuous.
C	Adjacent category	Relates two continuous variables; diet group is categorical.
D	Dropped element	Handles only two groups; multiple t-tests inflate Type I error.

8. Answer B Concept 5 • transfer

Which research question is best answered with a chi-square test of independence?

Target word: question that fits chi-square **Decides:** Both variables are categorical.

A	Wrong stem number	Three group means call for ANOVA.
B	Correct	Two categorical variables in a contingency table.
C	Adjacent category	Two group means call for a t-test.
D	Confusion set	Two continuous variables call for regression.

9. Answer D Concept 6

In a study of 250 adults ~~recruited through a fitness app~~, the correlation between daily steps and resting heart rate is $r = -.60$. Which interpretation is correct?

Target word: interpret r **Decides:** Sign gives direction; r^2 gives variance shared.

A	Inverted operation	The sign shows direction, not strength.
B	Dropped element	Skips squaring r .
C	Confusion set	Correlation does not establish cause.
D	Correct	More steps go with lower heart rate; $(-.60)^2 = .36$.

10. Answer B Concept 7

A hospital unit, ~~which renovated its lobby last spring~~, found that predicted weekly overtime hours = $12 + 0.8 \times$ (weekly admissions). How many overtime hours are predicted for a week with 30 admissions?

Target word: predicted y **Decides:** Plug $x = 30$ into the equation.

A	Inverted operation	Multiplies the intercept by x instead of adding.
B	Correct	$12 + 0.8 \times 30 = 12 + 24 = 36$.
C	Confusion set	Adds x directly, treating the slope as 1.
D	Dropped element	Leaves out the intercept of 12.

11. Answer B Concept 7 · transfer

Using predicted overtime hours = $12 + 0.8 \times$ (weekly admissions), how many admissions would predict 44 overtime hours?

Target word: solve for x **Decides:** $x = (44 - 12) / 0.8$.

A	Wrong stem number	The intercept.
B	Correct	$(44 - 12) / 0.8 = 32 / 0.8 = 40$.
C	Dropped element	$44 / 0.8$ ignores the intercept.
D	Inverted operation	Plugs 44 in as x instead of solving for x .

12. Answer A Concept 8

Birth weights at a regional hospital ~~that delivers about 4,000 babies a year~~ are roughly normal, with a mean of 3,400 g and a standard deviation of 500 g. About what percentage of babies weigh between 2,400 g and 4,400 g?

Target word: percentage within a range **Decides:** 2,400 and 4,400 are 2 SD from the mean.

A	Correct	$\pm 1,000 \text{ g} = \pm 2 \text{ SD}$; about 95% of a normal distribution.
B	Dropped element	Counts only one side of the mean.
C	Confusion set	That is within 1 SD (2,900 to 3,900 g).
D	Adjacent category	That is within 3 SD (1,900 to 4,900 g).

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	5	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	1	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Levels of measurement	Nominal = named categories; ordinal = ordered categories with unequal gaps; interval = equal gaps, no true zero; ratio = equal gaps and a true zero.	Treating ranked labels as interval data because they can be coded 1–4.	Response options that are ordered words, not measured amounts.
2	Center of skewed data	The median resists outliers; the mean is pulled toward the tail, so report the median for skewed data.	Reporting the mean for data with a big outlier.	One value far from the rest, such as incomes or home prices.
3	z-scores	$z = (\text{score} - \text{mean}) / \text{SD}$; a raw score = $\text{mean} + z \times \text{SD}$.	Forgetting to subtract the mean or divide by the SD.	A mean, an SD, and one score.
4	Significance decisions and errors	Reject the null when $p \leq \alpha$. A Type I error rejects a true null (false positive); a Type II error fails to reject a false null (false negative).	Reading p as the chance the treatment works, or swapping Type I and II.	A p-value next to α , or a wrong conclusion to classify.
5	Choosing the test	Two group means → t-test; three or more means → ANOVA; two categorical variables → chi-square; two continuous variables → correlation or regression.	Running a t-test on three groups or a t-test on counts.	How many groups and whether the outcome is a mean or a category.
6	Interpreting correlation	The sign of r gives direction, the size gives strength, and r^2 is the share of variance shared; correlation does not show cause.	Thinking a negative r is weak, or using r instead of r^2 for variance explained.	An r value followed by interpretation choices.
7	Using a regression equation	Predicted $y = \text{intercept} + \text{slope} \times x$; to find x for a target y, $x = (y - \text{intercept}) / \text{slope}$.	Dropping the intercept or plugging y in where x belongs.	An equation of the form $\hat{y} = a + bx$.
8	The empirical rule	In a normal distribution, about 68% of values lie within 1 SD of the mean, 95% within 2 SD, and 99.7% within 3 SD.	Matching the wrong percentage to the number of SDs.	A normal distribution with a range given in raw units.

Course scope (public description): Introduction to Statistics for Research covers descriptive and inferential statistics used to address research questions. Levels of measurement, central tendency and variability, probability, distributions, correlation, hypothesis testing, t-tests, analysis of variance, linear regression, and chi-square tests are examined, with a focus on their implications for research. Reading and interpreting graphically presented statistical data is included. Before taking this course, students should complete C955 Applied Probability and Statistics or an equivalent course.